

$$\begin{aligned} 180 &= \frac{180-A}{2} + \frac{180-C}{2} + x \\ 180 &= 180 + x - \frac{(A+C)}{2} \\ 180 &= 180 + x - 45 \\ x &= 45^\circ \end{aligned}$$

Qn15. (a) Let x be the middle integer. Then the other two integers are $x-1$ and $x+1$. Therefore $(x-1)(x)(x+1)+x = x[(x-1)(x+1)+1] = x(x^2-1+1) = x^3$ which is a cube as required.

(b) Suppose x and y are the grades Brenda obtained in the other two courses, then her overall average for the full year was

$$\begin{aligned} \frac{5(75)+4(80)+90+x+y}{12} &= 80 \\ x+y = 960-(5(75)+4(80)+90) &= 175. \\ x+y &= 175. \end{aligned}$$

The highest percentage grade she could obtain in either course is 100%.
Suppose $x = 100\%$, then $y = 175 - 100 = 75\%$ which is the minimum possible mark.

NMC 2016 0-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. List all ordered triples (a, b, c) of positive integers which satisfy $a + \frac{b}{c} = 11$ and $b + \frac{a}{c} = 14$.

Qn2. In triangle $ABC, AB = 3, BC = 4$, and $AC = 6$. If BC is extended through C to D so that $CD = BC$, what is AD?

Qn3. Ampeirwe's Convenience Centre buys a calculator for p shillings (where $p > 0$), raises its price by $n\%$, then reduces this new price by 20%. If the final price is 20% higher than p, what is the value of n?

Qn4. A function f is defined so that if n is an odd integer, then $f(n) = n - 1$ and if n is an even integer, then $f(n) = n^2 - 1$. For example, if $n = 15$, then $f(n) = 14$ and if $n = -6$, then $f(n) = 35$, since 15 is an odd integer and -6 is an even integer. Determine all integers n for which $f(f(n)) = 3$.

Qn5. A student takes three exams. The second has twice as many questions as the first, and the third has three times as many questions as the first. The student answers exactly 75% of the questions correctly on the first exam, exactly 81% on the second exam, and exactly 85% on the third exam. Out of all the questions on the three exams, what percentage did he answer wrongly?

Qn6. The price of a ticket to attend a football match was Shs.10, 000 but due to a sponsorship deal by a company to pay up a certain amount of the ticket price for each person attending the match it was reduced. The number of spectators increased by 50%, but the amount of money received only increased by 20%. How much was the new ticket price after reduction?

Qn7. A large container, labeled R, is partially filled with 4 litres of red paint. Another large container, labeled W, is partially filled with 5 litres of white paint. A small empty can is completely filled with red paint taken from R, and the contents of the can then emptied into W. After thorough mixing of the contents of W, the can is completely filled with some of this mixture from W, and the contents of the can then emptied into R. The ratio of red paint to white in R is now $3 : 1$. What is the size of the can, in litres?

Qn8. For what non-zero value of the radius, r, does the volume of a sphere equal its surface area?

Qn9. What is the tens digit of the smallest six-digit positive integer that is divisible by each of $10, 11, 12, 13, 14$, and 15?

Qn10. Determine all values of x for which $\left(2 \cdot 4^{x^2-3x}\right)^2 = 2^{x-1}$.

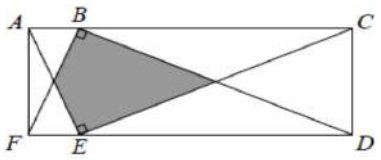
SECTION B

Qn11. My cat keeps to himself most of the time. I only heard him meow, hiss, and buzz on one out of the last 23 days. But I did hear him make at least one of these sounds each day. I heard him hiss on 6 days, buzz on 12 days, and meow on 15 days. On 2 days, I heard him meow and hiss but not buzz, and on 2 days, I heard him buzz and hiss but not meow. On how many days did I hear him meow and buzz but not hiss?

Qn12. Nankya and Kintu are driving on a circular track with radius 25 km. Nankya leaves the starting line first, going clockwise around the track at a speed of 80 km/h. Fifteen minutes after Nankya starts, Kintu leaves the same starting line, going counter- clockwise around the track at a speed of 100 km/h. For how many hours will Kintu have been driving when the two of them pass each other for the fourth time?

Qn13. In a psychology experiment, an image of a cat or an image of a dog is flashed briefly onto a screen and then Apio is asked to guess whether the image showed a cat or a dog. This process is repeated a large number of times with an equal number of images of cats and images of dogs shown. If Apio is correct 95% of the time when she guesses "dog" and 90% of the time when she guesses "cat", determine the ratio of the number of times she *guessed* "dog" to the number of times she *guessed* "cat".

Qn14. In the diagram, $ACDF$ is a rectangle with $AC = 200$ and $CD = 50$. Also, $\triangle FBD$ and $\triangle AEC$ are congruent triangles which are right-angled at B and E, respectively. What is the area of the shaded region?



Qn15. A formula for the sum of the first n positive integers is $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

(a) What is the sum of the first 200 positive integers, $1 + 2 + 3 + \cdots + 198 + 199 + 200$?

(b) Calculate the sum of the 50 consecutive integers beginning at 151, that is,

$$151 + 152 + 153 + \cdots + 198 + 199 + 200.$$

(c) Starting with the sum of the first 1000 positive integers, $1 + 2 + 3 + \cdots + 999 + 1000$, every third integer is removed to create the new sum

$$1 + 2 + 4 + 5 + 7 + 8 + 10 + 11 + \cdots + 998 + 1000.$$

Calculate this new sum.

NMC 2016 O-LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1.

$$a + \frac{b}{c} = 11 \Longrightarrow c = \frac{b}{11 - a} \tag{1}$$

$$b + \frac{a}{c} = 14 \Longrightarrow c = \frac{a}{14 - b} \tag{2}$$

Equating the two equations (1) and (2)

$$\frac{b}{11 - a} = \frac{a}{14 - b} \quad or \quad (11 - a)a = (14 - b)b$$

Clearly $a < 11$ and $b < 14$ since $a, b, \& \ c$ are positive integers.

By taking $a \in \{1, 2, \cdots, 10\}$, the solution set for the LHS is $\{10, 18, 24, 30\}$. Likewise, the solution set for the RHS is $\{13, 24, 33, 40, 48, 49\}$

The common solution is 24, which happens when $a = 8$ or $a = 3$ and $b = 12$ or $b = 2$.

Case 1 For $a = 3, b = 2$ will make c in Eqn (2) non integer.

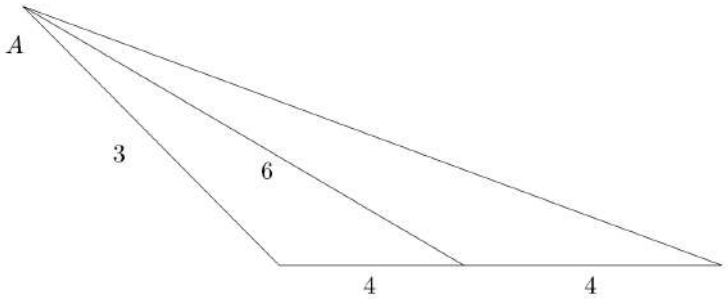
Case 2 For $a = 3, b = 12$ will make c in Eqn (2) non integer.

Case 3 For $a = 8, b = 2$ will make c in Eqn (2) non integer.

For $a = 8, b = 12$ gives an integer value of $c = 4$. Thus the solution set

$$(a, b, c) = (8, 12, 4)$$

Qn2.



By the Cosine rule

$$6^2 = 3^2 + 4^2 - 24\cos B \quad \Rightarrow \quad \cos B = -\frac{11}{24}$$

Similarly,

$$\begin{aligned} AD^2 &= 3^2 + 8^2 - 24\cos B \\ AD^2 &= 3^2 + 8^2 - 24\left(-\frac{11}{24}\right) \\ AD^2 &= 73 + 22 = 95 \\ &\Leftrightarrow \quad AD = \sqrt{95} \end{aligned}$$

Qn3. When the price of p shillings is raised by $n\%$, the price is multiplied by $1 + \frac{n}{100}$

When the new price is reduced by 20%, the new price is multiplied by $1 - \frac{20}{100} = \frac{80}{100}$ Therefore, after these two price adjustments, the price is

$$p\left(1 + \frac{n}{100}\right)\left(\frac{80}{100}\right)$$

We are told that this final price is 20% higher than p, and so the final price equals

$$p\left(1 + \frac{20}{100}\right) \quad \text{or} \quad p\left(\frac{120}{100}\right)$$

In other words,

$$p\left(1+\frac{n}{100}\right)\left(\frac{80}{100}\right)=p\left(\frac{120}{100}\right)$$

Simplifying and using the fact that $p \neq 0$, we obtain

$$80\left(1+\frac{n}{100}\right)=120$$

Thus,

$$1+\frac{n}{100}=\frac{120}{80}=\frac{3}{2}=\frac{150}{100}$$

and so

$$\frac{n}{100}=\frac{50}{100} \quad \text{or} \quad n=50$$

Qn4. *Solution 1*

Let $m = f(n)$. The equation $f(f(n)) = 3$ becomes $f(m) = 3$.

Suppose that $f(m) = 3$ and m is odd. By definition, we have $f(m) = m - 1 = 3$ and so $m = 4$, which is not odd, so this case cannot happen.

Suppose that $f(m) = 3$ and m is even. By definition, we have $f(m) = m^2 - 1 = 3$ and so $m^2 = 4$ or $m = \pm 2$, each of which is even.
Therefore, if $f(f(n)) = 3$, then $f(n) = 2$ or $f(n) = -2$.

Suppose that $f(n) = 2$ or $f(n) = -2$ and n is odd. By definition, we have $n-1 = 2$ (giving $n = 3$) or $n-1 = -2$ (giving $n = -1$). Each of these resulting values of n is odd.

Suppose that $f(n) = 2$ or $f(n) = -2$ and n is even. Then $n^2 - 1 = 2$ or $n^2 - 1 = -2$ which give $n^2 = 3$ or $n^2 = -1$, neither of which is possible if n is an integer. Thus, the integers n for which $f(f(n)) = 3$ are $n = 3$ and $n = -1$. (We can check by substitution that each of these satisfies the original equation.)

Solution 2

We consider the cases of n even and n odd separately.

Suppose that n is even. Then n^2 is even and so $f(n) = n^2 - 1$ must be odd. Thus, $f(f(n)) = f(n^2 - 1) = (n^2 - 1) - 1 = n^2 - 2$, since $f(m) = m - 1$ when m is odd.
For n to be even and $f(f(n)) = 3$, we must have $n^2 - 2 = 3$ or $n^2 = 5$. There are no integer solutions to this equation, and so there are no solutions in this case.

Suppose that n is odd. Then $f(n) = n - 1$ must be even. Thus, $f(f(n)) = f(n - 1) = (n - 1)^2 - 1 = n^2 - 2n + 1 - 1 = n^2 - 2n$. For n to be odd and $f(f(n)) = 3$, we must have $n^2 - 2n = 3$ or $n^2 - 2n - 3 = 0$. Factoring, we obtain $(n - 3)(n + 1) = 0$ and so $n = 3$ or $n = -1$, both of which are odd.

Thus, the integers n for which $f(f(n)) = 3$ are $n = 3$ and $n = -1$. (We can check by substitution that each of these satisfies the original equation.)

Qn5. Let the number of equations in the first exam be n, then the student answered correctly

$$\begin{aligned} & 0.75n + 0.81 \times 2n + 0.85 \times 3n \quad \text{out of} \quad (n + 2n + 3n) \\ \Rightarrow \quad & 0.75n + 0.81 \times 2n + 0.85 \times 3n \quad \text{out of} \quad 6n \end{aligned}$$

Those answered correctly

$$\frac{0.75n + 0.81 \times 2n + 0.85 \times 3n}{6n} = \frac{0.75 + 0.81 \times 2 + 0.85 \times 3}{6} = 0.82 = 82\%$$

Thus 82% answered correctly, and $100 - 82 = 18\%$ answered wrongly.

Qn6. If the new price ticket is x and the original number of spectators was N, then

$$1.5 \times Nx = 1.2 \times 10,000 \times N \quad \text{the increase in revenue by 2\%}$$

$$\Rightarrow \quad x = \frac{1.2(10,000N)}{1.5N} = 8,000$$

is the new ticket price.

Qn7. Let x denote the size of the can. After the first transfer, container R has

$$4 - x \text{ Red}$$

And container W has

$$x \text{ Red} \quad , \quad 5 \text{ White}$$

After the second transfer, R has

$$\left((4 - x) + x\frac{x}{5 + x}\right) \text{ Red} \quad , \quad x\frac{x}{5 + x} \text{ White}$$

Thus using the ratio 1 : 3

$$\begin{aligned} (4 - x)(5 + x) + x^2 &= 3(5x) \\ 20 &= 16x \end{aligned}$$

$$\Rightarrow \quad x = \frac{20}{16} = \frac{5}{4} \text{ litres}$$

Qn8. For the volume of a sphere equalling its surface area $\frac{4}{3}\pi r^3 = 4\pi r^2$.

$$\frac{4}{3}\pi r^3 - 4\pi r^2 = 0$$

$$4\pi r^2 \left(\frac{1}{3}r - 1\right) = 0$$

Thus the non zero r value is $r = 3$.

Qn9. Among the list $10, 11, 12, 13, 14, 15$, the integers 11 and 13 are prime.
Also, $10 = 2 \times 5$ and $12 = 2 \times 2 \times 3$ and $14 = 2 \times 7$ and $15 = 3 \times 5$.
For an integer N to be divisible by each of these six integers, N must include at least two factors of 2 and one factor each of $3, 5, 7, 11, 13$.

Note that

$$2^2 \times 3 \times 5 \times 7 \times 11 \times 13 = 60060.$$

(This is the least common multiple of $10, 11, 12, 13, 14, 15$.)

To find the smallest six-digit positive integer that is divisible by each of $10, 11, 12, 13, 14, 15$, we can find the smallest six-digit positive integer that is a multiple of 60 060.

Note that $1 \times 60\ 060 = 60\ 060$ and that $2 \times 60\ 060 = 120\ 120$. Therefore, the smallest six-digit positive integer that is divisible by each of $10, 11, 12, 13, 14, 15$ is 120 120. Therefore, the tens digit of this number is 2.

Qn10. Using exponent laws, we obtain the following equivalent equations:

$$\begin{aligned}
\left(2 \cdot 4^{x^2-3x}\right)^2 &= 2^{x-1} \\
2^2 \cdot 4^{2(x^2-3x)} &= 2^{x-1} \\
2^2 \cdot (2^2)^{2x^2-6x} &= 2^{x-1} \\
2^2 \cdot 2^{2(2x^2-6x)} &= 2^{x-1} \\
\frac{2^2 \cdot 2^{4x^2-12x}}{2^{x-1}} &= 1 \\
2^{2+4x^2-12x-(x-1)} &= 1 \\
2^{4x^2-13x+3} &= 1
\end{aligned}$$

Since $2^0 = 1$, this last equation is true exactly when $4x^2 - 13x + 3 = 0$ or $(4x-1)(x-3) = 0$. Therefore, $x = \frac{1}{4}$ or $x = 3$. We can check by substitution that each of these values of x satisfies the original equation.

SECTION B

Qn11. Let M denote the days on which the cat meowed, H the days on which he hissed, and P the days on which he buzzed. Given the data, the following Venn diagram represents the situation, where the expressions inside the circles are the cardinalities of disjoint sections:

We are looking for x. The labeled sections of the Venn diagram are disjoint sets, so they should add to 23. That is,

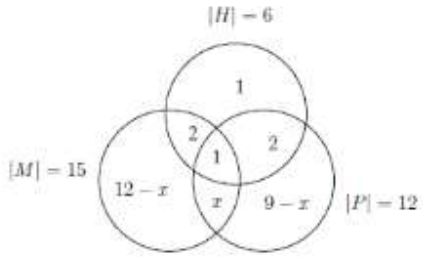
$$23 = 1 + 1 + 2 + 2 + x + (9 - x) + (12 - x) = 27 - x$$

Therefore, $x = 4$.

Qn12. Since the track is circular with radius 25 km, then its circumference is $2\pi(25) = 50\pi$ km.

In the 15 minutes that Nankya drives at 80 km/h, he drives a distance of $\frac{1}{4}(80) = 20$ km (because 15 minutes is one-quarter of an hour).

When Kintu starts driving, she drives in the opposite direction to Nankya. Suppose that Nankya and Kintu



meet for the first time after Kintu has been driving for t hours. During this time, Kintu drives at 100 km/h, and so drives $100t$ km. During this time, Nankya drives at 80 km/h, and so drives $80t$ km.

Since they start $50\pi - 20$ km apart along the track (the entire circumference minus the 20 km that Nankya drove initially), then the sum of the distances that they travel is $50\pi - 20$ km. Therefore, $100t + 80t = 50\pi - 20$ and so $180t = 50\pi - 20$ or $t = \frac{5\pi-2}{18}$.

Suppose that Nankya and Kintu meet for the next time after an additional T hours. During this time, Kintu drives $100T$ km and Nankya drives $80T$ km. In this case, the sum of the distances that they drive is the complete circumference of the track, or 50π km. Thus, $180T = 50\pi$ or $T = 5\pi 18$.

The length of time between the first and second meetings will be the same as the amount of time between the second and third, and between the third and fourth meetings. Therefore, the total time that Kintu has been driving when she and Nankya meet for the fourth time will be

$$t + 3T = \frac{5\pi - 2}{18} + 3 \cdot \frac{5\pi}{18} = \frac{20\pi - 2}{18} = \frac{10\pi - 1}{9} \quad \text{hours}$$

Qn13. Suppose that Opio guesses "cat" c times and guesses "dog" d times.

When she guesses "dog", she is correct 95% of the time.
When she guesses "cat", she is correct 90% of the time.

Thus, when she guesses "cat", she is shown $0.9c$ images of cats and so $c - 0.9c = 0.1c$ images of dogs.

Thus, when she guesses "dog", she is shown $0.95d$ images of dogs and so $d - 0.95d = 0.05d$ images of cats.

(We assume that c and d have the property that $0.9c$ and $0.95d$ are integers.) Therefore, the total number of images of cats that she is shown is $0.9c + 0.05d$ and the total number of images of dogs that she is shown is $0.1c + 0.95d$.

But the number of images of cats equals the number of images of dogs.
Thus, $0.9c + 0.05d = 0.1c + 0.95d$, which gives $0.8c = 0.9d$ or $\frac{d}{c} = \frac{0.8}{0.9}$

Therefore, the ratio of the number of times that she guessed "dog" to the the number of times that she guessed "cat" is $8 : 9$.

Qn14. Since $\triangle FBD$ is congruent to $\triangle AEC$, then $FB = AE$. Since $\triangle FAB$ and $\triangle AFE$ are each right-angled, share a common side AF and have equal hypotenuses ($FB = AE$), then these triangles are congruent, and so $AB = FE$.

Now $BAFE$ has two right angles at A and F (so AB and FE are parallel) and has equal sides $AB = FE$ so must be a rectangle. This means that $BCDE$ is also a rectangle.

Now the diagonals of a rectangle partition it into four triangles of equal area. (Diagonal AE of the rectangle

splits the rectangle into two congruent triangles, which have equal area. The diagonals bisect each other, so the four smaller triangles all have equal area.)

Since $\frac{1}{4}$ of rectangle $ABEF$ is shaded and $\frac{1}{4}$ of rectangle $BCDE$ is shaded, then $\frac{1}{4}$ of the total area is shaded. (If the area of $ABEF$ is x and the area of $BCDE$ is y, then the total shaded area is $\frac{1}{4}x + \frac{1}{4}y$, which is $\frac{1}{4}$ of the total area $x + y$.)

Since $AC = 200$ and $CD = 50$, then the area of rectangle $ACDF$ is $200(50) = 10000$, so the total shaded area is $\frac{1}{4}(10000) = 2500$.

Qn15. (a) Using the formula given, the sum of the fist 200 positive integers

$$1 + 2 + 3 + \cdots + 198 + 199 + 200 = \frac{200(200 + 1)}{2} = 100(201) = 20100$$

(b) The sum of the fist 200 positive integers is equal to the sum of the fist 150 positive integers added to the sum of the 50 consecutive integers $151 + 152 + 153 + \cdots + 198 + 199 + 200$.

That is,

$$1 + 2 + \cdots + 198 + 199 + 200 = (1 + 2 + \cdots + 148 + 149 + 150) + (151 + 152 + \cdots + 198 + 199 + 200)$$

Therefore, the sum of the 50 consecutive integers $151 + 152 + 153 + \cdots + 198 + 199 + 200$ is equal to the difference between the sum of the fist 200 positive integers and the sum of the fist 150 positive integers. From part (15a), the sum of the fist 200 positive integers is 20100.

Using the formula, the sum of the fist 150 positive integers

$$1 + 2 + 3 + \cdots + 148 + 149 + 150 = \frac{150(150 + 1)}{2} = 10075(151) = 11325$$

Therefore, the sum of the 50 consecutive integers beginning at 151,

$$\begin{aligned} 151 + 152 + \cdots + 199 + 200 &= (1 + 2 + \cdots + 199 + 200) - (1 + 2 + \cdots + 149 + 150) \\ &= 20100 - 11325 \\ &= 8775 \end{aligned}$$

(c) Let the required sum, $1 + 2 + 4 + 5 + 7 + 8 + 10 + 11 + \cdots + 998 + 1000$, be S.

Let the sum of the first 333 positive multiples of 3, $3 + 6 + 9 + 12 + \cdots + 996 + 999$, be T.

The sum of the first 1000 positive integers is equal to $S + T$. That is,

$$1 + 2 + \cdots + 998 + 999 + 1000 = (1 + 2 + 4 + 5 + 7 + \cdots + 998 + 1000) + (3 + 6 + \cdots + 996 + 999)$$

Therefore, the required sum S is equal to the difference between the sum of the first 1000 positive integers and T.

Using the formula, the sum of the first 1000 positive integers

$$1 + 2 + 3 + \cdots + 998 + 999 + 1000 = \frac{1000(1000 + 1)}{2} = 500(1001) = 500500$$

We may determine the sum T by first removing the common factor 3 from each term and then using the formula.

$$\begin{aligned} 3+6+9+\cdots+993+996+999 &= 3(1+2+3+\cdots+331+332+333) \\ &= 3\left(\frac{3333(333+1)}{2}\right) \\ &= 3(333\times 167) \\ &= 166833 \end{aligned}$$

Finally, the required sum S is $500500 - 166833 = 333667$.